

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_11apr25.parquet

### BiLSTM Multi-Step Training Report

Report Generated: 2025-05-07 08:54:36

Training Start: 2025-05-06 20-12-54

Training End: 2025-05-06 20-25-30

Training Duration: 12.60 minutes

#### Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.6124

MSE: 4.8485

RMSE: 2.2019

R2 Score: 0.1517

#### Multi-Step Forecast:

Forecast Steps: 20

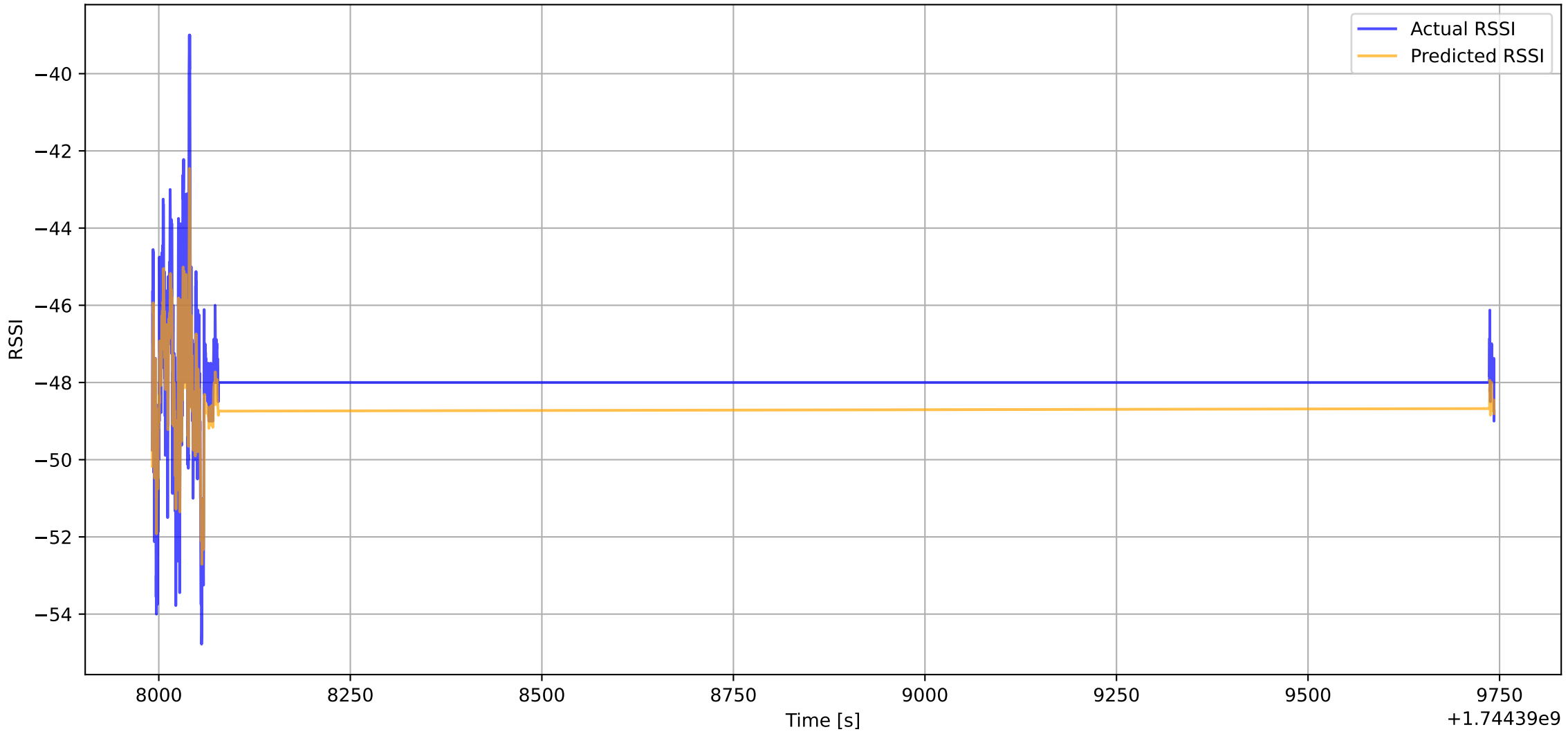
MAE: 0.0385

MSE: 0.0024

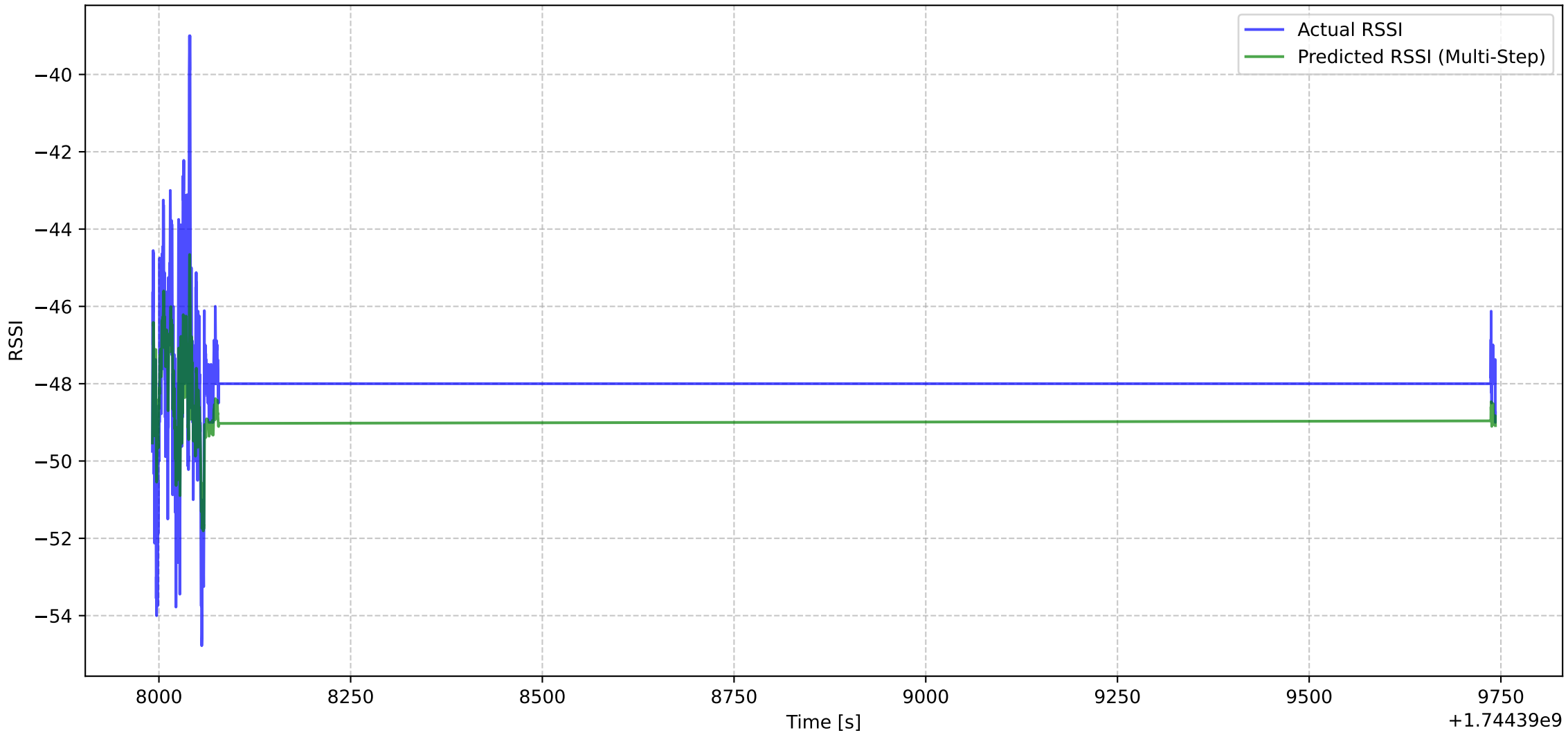
RMSE: 0.0493

R2 Score: 0.6986

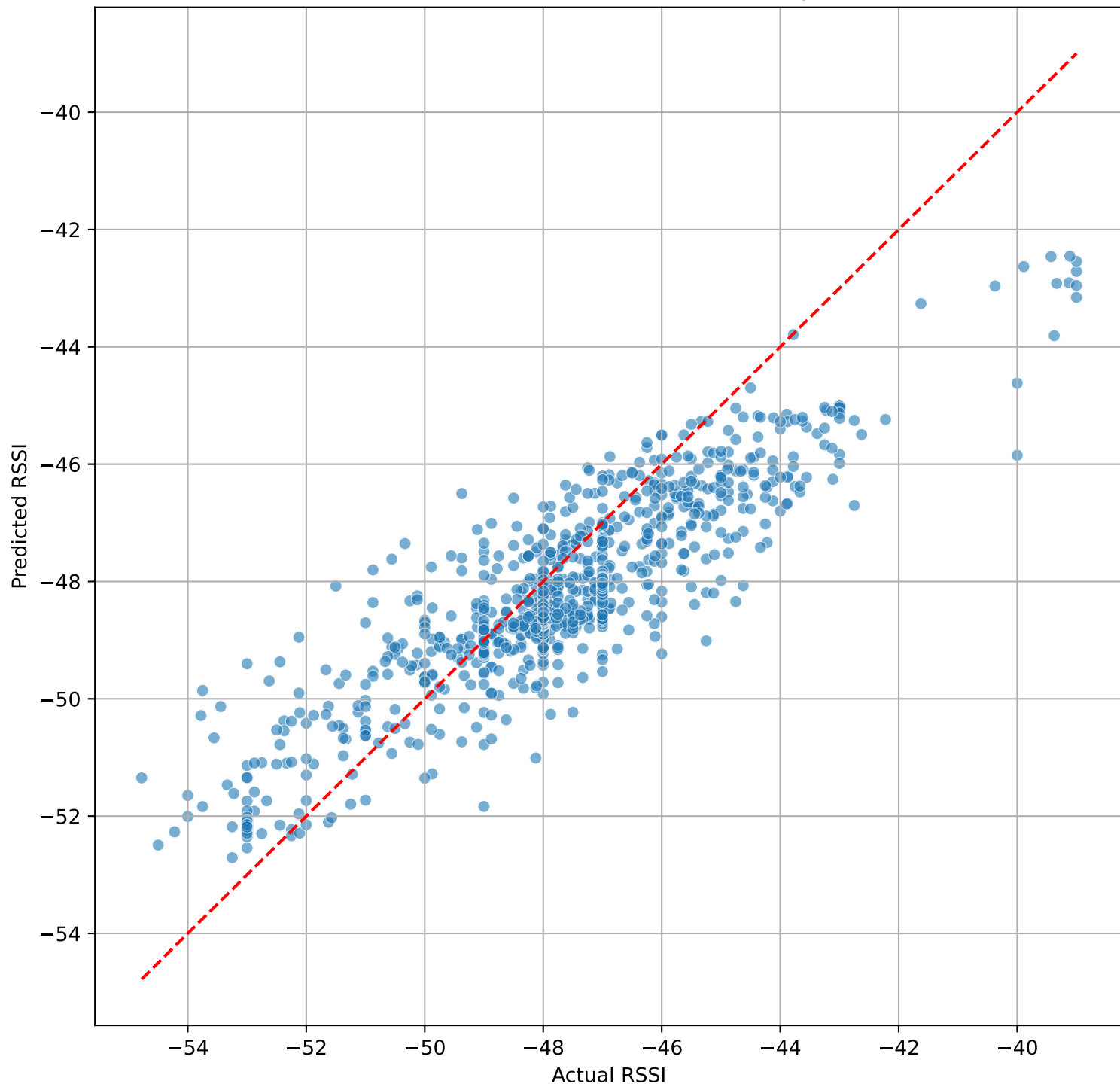
Actual vs Predicted RSSI (First Step)



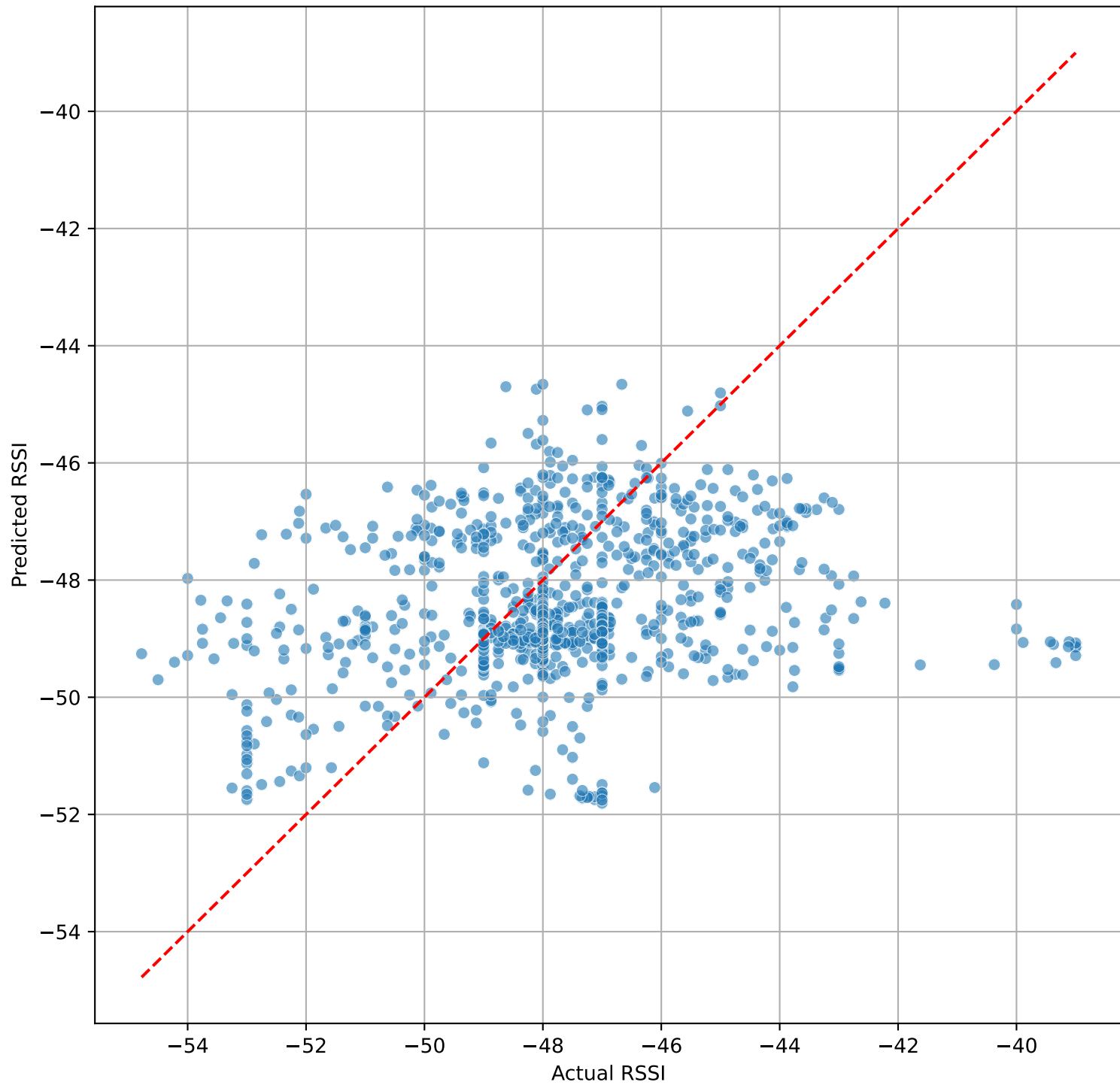
Actual vs Predicted RSSI (Multi-Step (step 20))



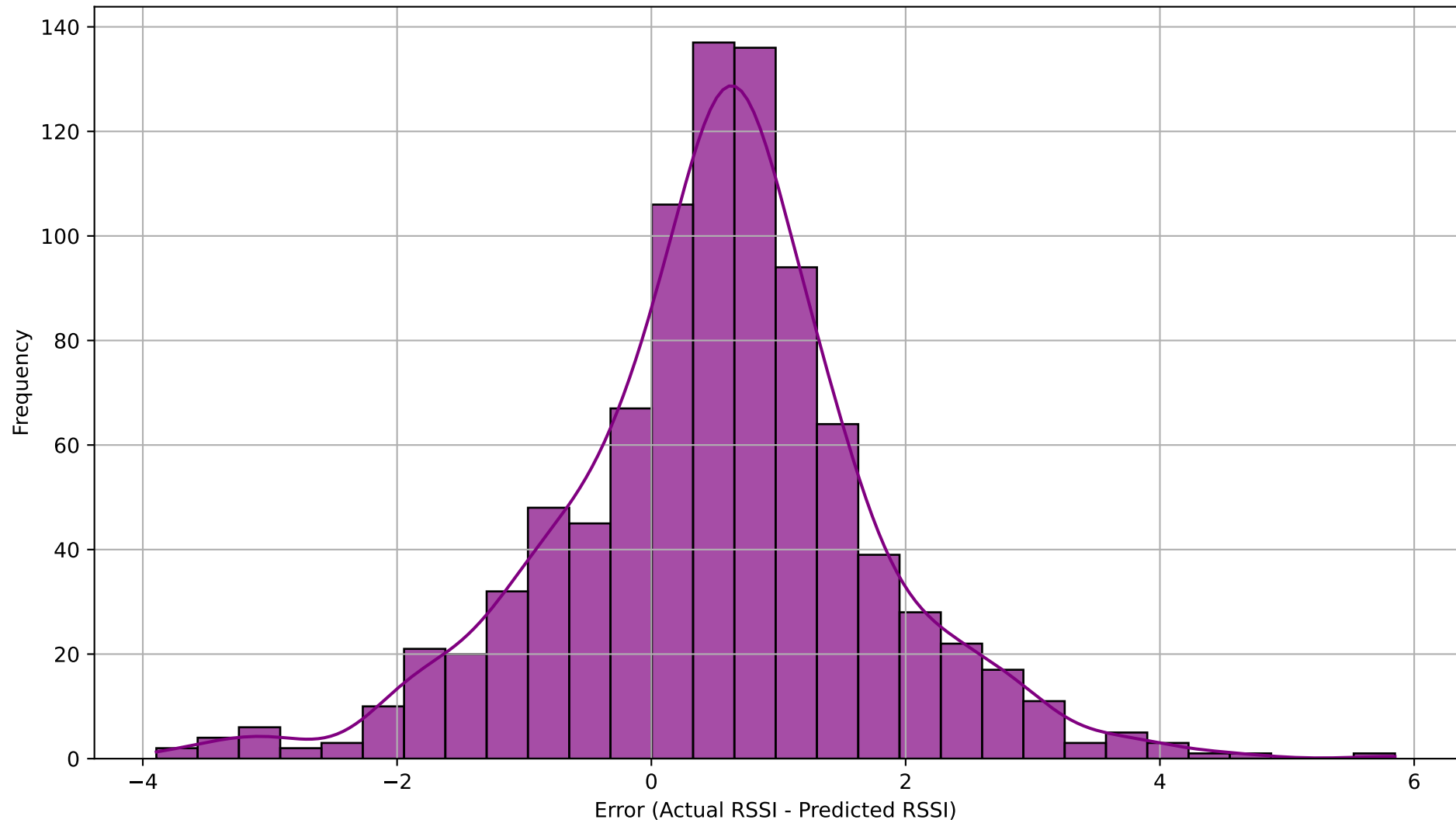
Actual vs Predicted RSSI (First Step)



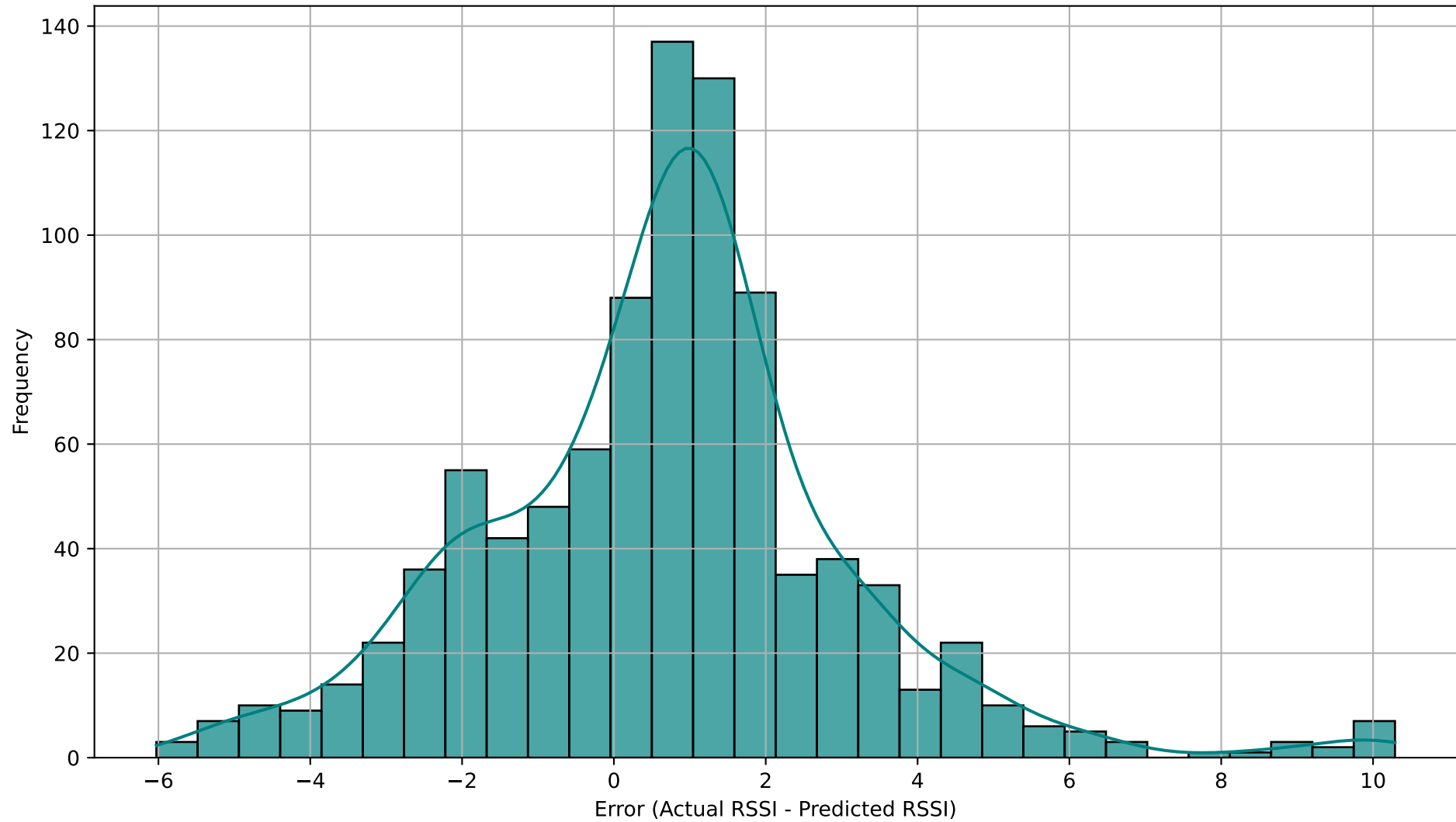
Actual vs Predicted RSSI (Last Step - step 20)



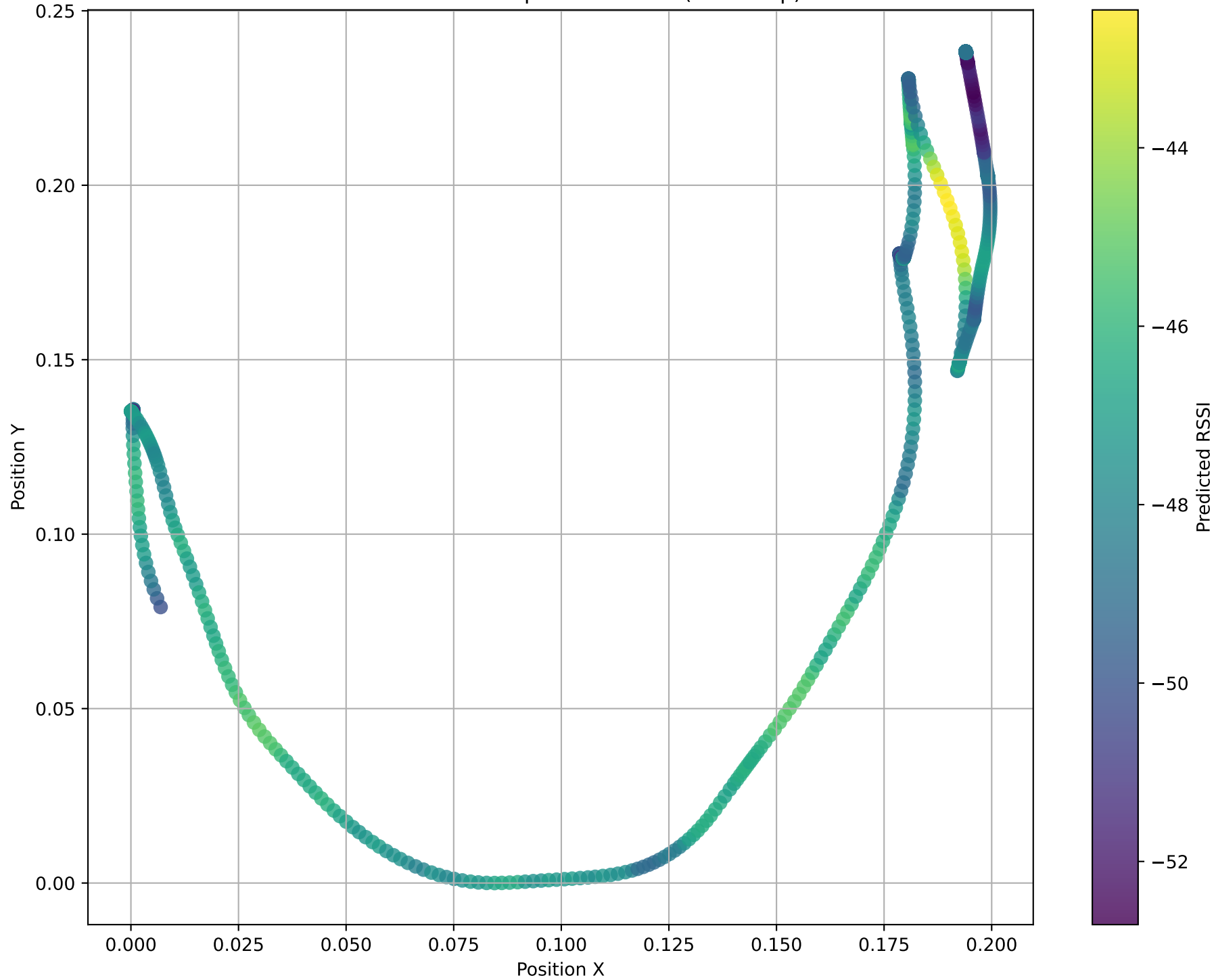
Distribution of Prediction Errors (First Step)



Distribution of Prediction Errors (Last Step - step 20)

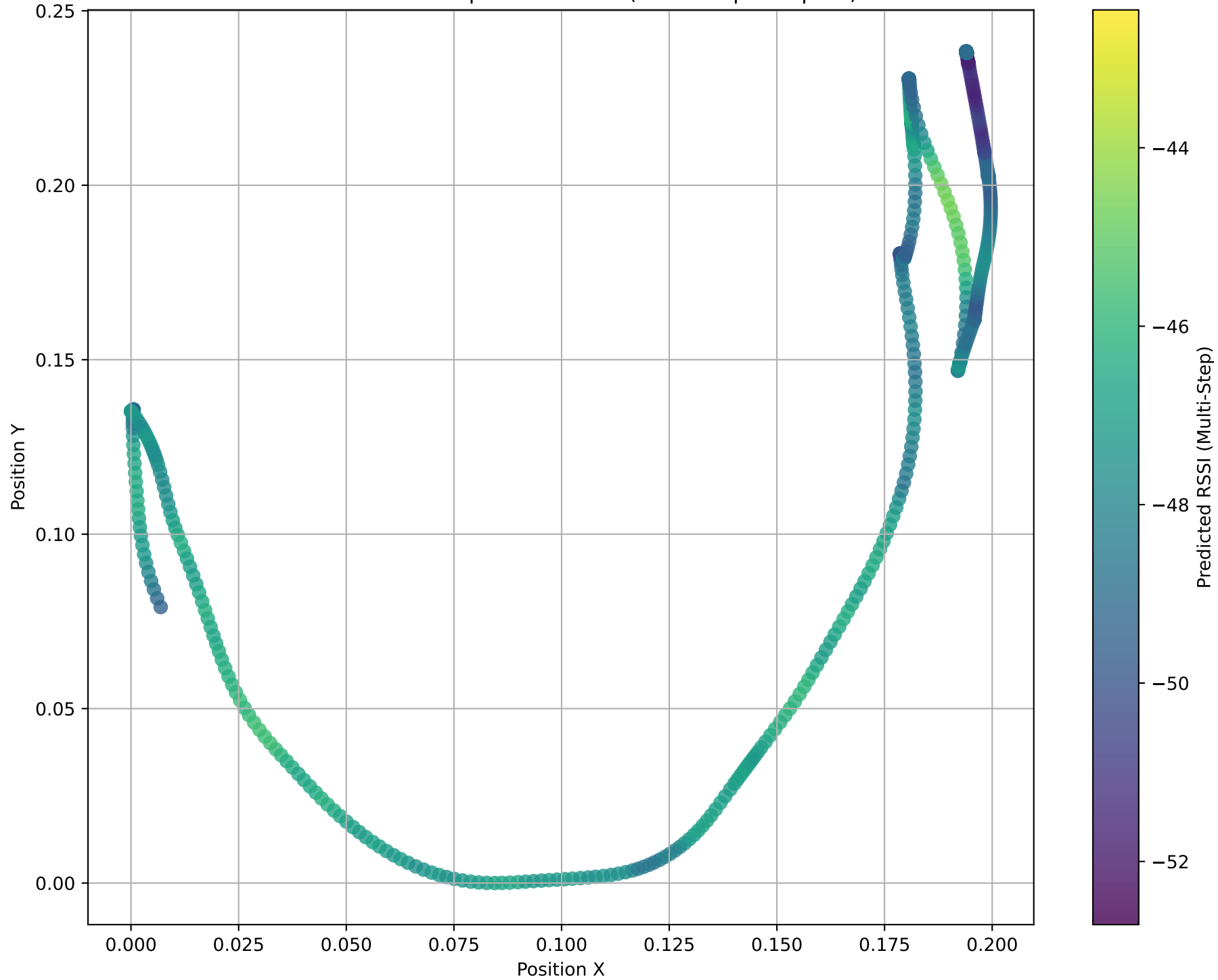


Predicted RSSI in Spatial Context (First Step)

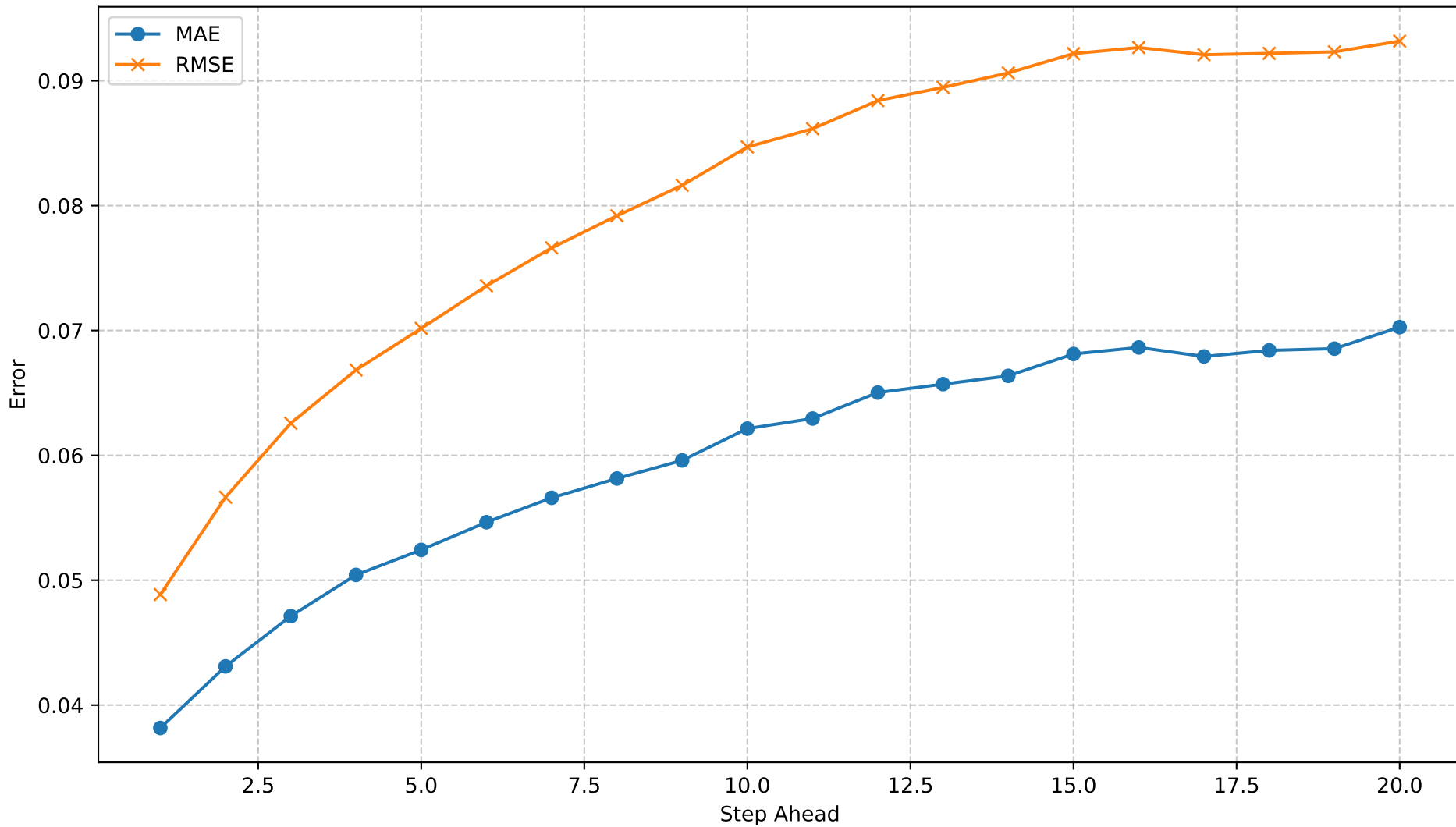




Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



### Resumo Automático

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Dataset: rssi\_odom\_dataset\_11apr25.parquet

Modelo: BiLSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.04 / 0.05
- Mediana : 0.06 / 0.09
- 20º passo : 0.07 / 0.09

Crescimento do erro: 84.1% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.07), 16 (0.07), 19 (0.07)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.